

WEEK 1 REPORT

DATA CLEANING AND TRANSFORMATION REPORT

SUPERSTORE SALES DATA IMPORT, POWER QUERY & DATA CLEANING

Introduction

Microsoft Power BI is a powerful data visualization and data analysis tool used to prepare data and create meaningful reports and dashboards. In this task, I used **Power BI Desktop** and **Power Query Editor** to import and prepare the **Superstore Sales Dataset**.

The dataset was examined for correct data types, missing values, duplicate or unique records, and unwanted spaces or characters in text fields. Various transformation techniques such as renaming columns, splitting columns, and formatting text were performed. I also created useful date-based columns such as **Year, Quarter, Month, Day, and Day of Week** from the **Order Date** column. After completing the cleaning and transformation process, the data was validated and loaded into the Power BI model.

Objectives

- To load the Superstore Sales Dataset into Power BI.
- To understand and explore Report, Data, and Model views.
- To examine the dataset using Power Query Editor.
- To verify and modify incorrect data types.
- To identify blank, null, empty, and error values.
- To check the uniqueness of records.
- To clean text values using Trim and Clean functions.
- To rename and transform the required columns.
- To split columns wherever necessary.
- To generate Year, Quarter, Month, Day, and Day of Week from Order Date.
- To prepare the final cleaned dataset for analysis in Power BI.

Tools and Dataset :

Tool Used: Microsoft Power BI Desktop

Data Preparation: Power Query Editor

Dataset: Superstore Sales Dataset

File Format: CSV

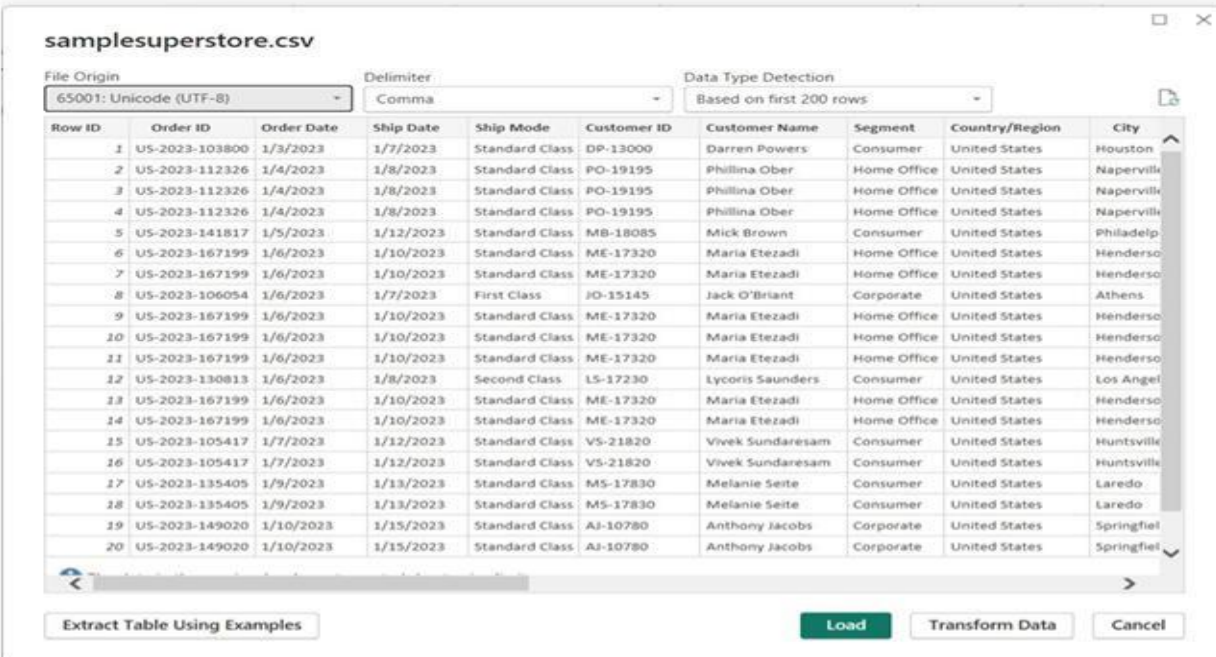
Power Query Editor was used as the main data preparation environment to inspect, clean, transform, and organize the Superstore Sales data before using it for report creation.

1. Importing the Dataset :

Initially, I opened Power BI Desktop and imported the Superstore Sales CSV file. After selecting the dataset, I opened it in Power Query Editor to perform the required data cleaning and transformation activities.

Steps Followed :

1. Launched **Power BI Desktop**.
2. Selected the **Home** tab.
3. Clicked on **Get Data**.
4. Selected **Text/CSV** from the available options.
5. Browsed and selected the Superstore Sales Dataset.
6. Clicked **Open**.
7. Verified the displayed data preview.
8. Selected **Transform Data**.
9. The dataset was opened in **Power Query Editor** for further processing.



The screenshot shows the Power Query Editor interface with the 'samplesuperstore.csv' dataset loaded. The 'File Origin' is set to '65001: Unicode (UTF-8)', the 'Delimiter' is 'Comma', and 'Data Type Detection' is 'Based on first 200 rows'. The table preview shows the following data:

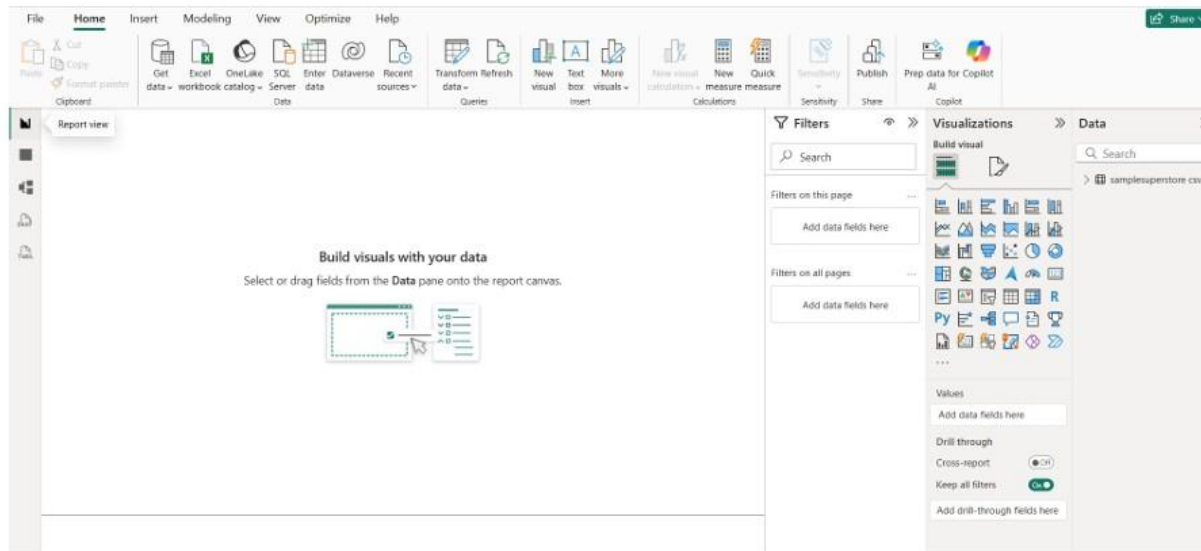
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelph
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
8	US-2023-106054	1/6/2023	1/7/2023	First Class	JO-15145	Jack O'Briant	Corporate	United States	Athens
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
12	US-2023-130813	1/6/2023	1/8/2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Angel
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville
17	US-2023-135405	1/9/2023	1/13/2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo
18	US-2023-135405	1/9/2023	1/13/2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo
19	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfiel
20	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfiel

2. Exploring Power BI Views

Power BI Desktop provides different views that help in analyzing and managing data.

Report View

Report View is mainly used to design dashboards and reports by adding charts, tables, cards, and other visual elements.



Data View

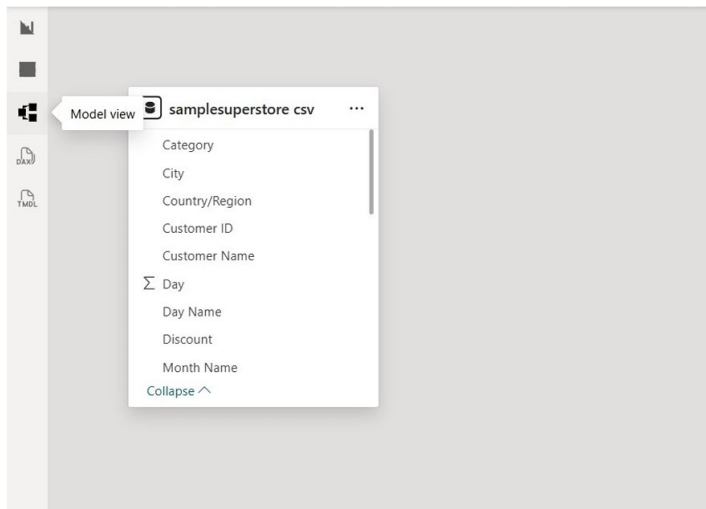
Data View helps to examine the actual records available in the dataset. It allows the user to inspect individual rows and columns.

The screenshot shows the Power BI Desktop interface in Data View. The top ribbon is the same as in the Report View. The main canvas displays a table of data from the 'samplesuperstore.csv' dataset. The table has columns for Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, City, State/Province, and Postal Code. The data is sorted by Order ID in descending order. The table contains 20 rows of data, showing various orders and their details.

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code
5-2023-130813	06 January 2023	08 January 2023	Second Class	LS-17230	Lyctoris Saunders	Consumer	United States	Los Angeles	California	90049
US-2023-148614	20 January 2023	25 January 2023	Standard Class	MV-17485	Mark Van Huff	Consumer	United States	Los Angeles	California	90049
US-2023-164903	20 February 2023	24 February 2023	Standard Class	SR-20740	Steven Roelle	Home Office	United States	Los Angeles	California	90049
US-2023-102652	06 April 2023	12 April 2023	Standard Class	AV-10555	Andy Votiv	Corporate	United States	Los Angeles	California	90049
US-2023-102652	06 April 2023	12 April 2023	Standard Class	AV-10555	Andy Votiv	Corporate	United States	Los Angeles	California	90049
US-2023-120740	15 April 2023	15 April 2023	Same Day	PS-18970	Paul Stevenson	Home Office	United States	Los Angeles	California	90049
US-2023-110849	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049
US-2023-110849	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049
US-2023-110849	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049
US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049
US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049
US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049
US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049
US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049
US-2023-106439	31 October 2023	04 November 2023	Standard Class	OG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049
US-2023-106439	31 October 2023	04 November 2023	Standard Class	OG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049
US-2023-106439	31 October 2023	04 November 2023	Standard Class	OG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049
US-2023-106439	31 October 2023	04 November 2023	Standard Class	OG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049
US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049
US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049
US-2023-153619	03 November 2023	03 November 2023	Same Day	TC-21535	Tracy Collins	Home Office	United States	Los Angeles	California	90049
US-2023-138737	07 December 2023	10 December 2023	First Class	FP-14320	Frank Preis	Consumer	United States	Los Angeles	California	90049
US-2023-113579	12 December 2023	15 December 2023	Second Class	KD-16345	Katherine Douch	Consumer	United States	Los Angeles	California	90049
US-2024-138219	23 March 2024	30 March 2024	Standard Class	BP-11095	Bart Pistole	Corporate	United States	Los Angeles	California	90049
US-2024-121188	28 August 2024	04 September 2024	Standard Class	CB-12025	Cassandra Brandow	Consumer	United States	Los Angeles	California	90049

Model View

Model View is used to understand the structure of the data model and manage relationships between different tables.



Steps Followed :

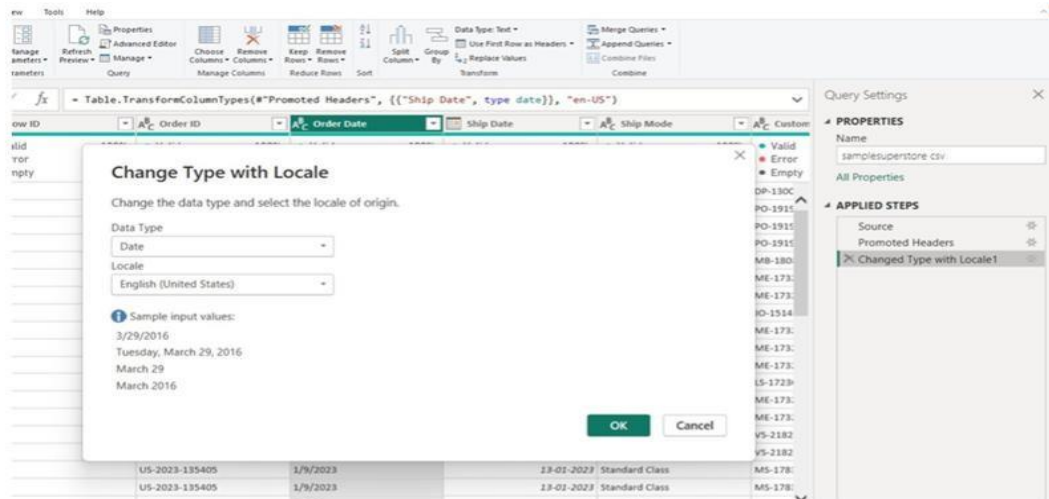
1. Opened the Power BI Desktop application.
 2. Viewed the **Report View**.
 3. Opened the **Data View** to inspect the dataset.
 4. Checked the **Model View**.
 5. Reviewed the table structure and available fields.
 6. Confirmed that the dataset was loaded correct.
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3. Verifying and Modifying Data Types

After importing the dataset, I verified the data type assigned to each important column. Correct data types are necessary to perform accurate calculations and analysis.

Steps Followed :

1. Opened the dataset in **Power Query Editor**.
2. Selected the required column.
3. Checked the data type displayed beside the column name.
4. Identified columns that required data type changes.
5. Used **Data Type → Change Type** to modify them.
6. Set categorical fields to **Text**.
7. Set quantity-related fields to **Whole Number**.
8. Set sales and profit fields to **Decimal Number**.
9. Changed the **Order Date** field to **Date**.



4. Cleaning Text Data Using Trim and Clean

Text fields may contain unnecessary spaces or unwanted characters that can affect data analysis. Therefore, I cleaned the categorical columns using the **Trim** and **Clean** options.

The main categorical columns considered were:

- Category
- Sub-Category
- Segment

Steps Followed :

1. Selected the **Category** column.
2. Opened the **Transform** tab.
3. Selected **Format**.
4. Applied the **Trim** operation.
5. Checked the values after removing unnecessary spaces.
6. Again selected **Transform** → **Format** → **Clean**.
7. Verified the cleaned values.
8. Applied the same process to **Sub-Category**.
9. Repeated the cleaning process for **Segment**.
10. Reviewed the columns to ensure consistent text formatting.

Trim helps remove unwanted spaces from the beginning and end of text values, while **Clean** removes non-printable characters.

Product ID	Category	Sub-Category	Product Name	Sales
OFF-PA-10000174	Office Supplies	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets...	100%
OFF-BI-10004094	Office Supplies	Binders	Gbc Standard Plastic Binding Systems Combs	100%
OFF-LA-10000123	Office Supplies	Labels	Avery 508	100%
OFF-ST-100002743	Office Supplies	Storage	Safco Boltless Steel Shelving	100%
OFF-AB-100001478	Office Supplies	Art	Avery Hi-Liter Everbold Pen Style Fluorescent Highlighters, 4/Pack	100%
FUR-CH-10004063	Furniture	Chairs	Global Deluxe High-Back Manager's Chair	100%
OFF-AB-10001862	Office Supplies	Art	Rogers Handheld Barrel Pencil Sharpener	100%
OFF-AB-10000299	Office Supplies	Art	Dixon Prang Watercolor Pencils, 10-Color Set With Brush	100%
OFF-BI-10004652	Office Supplies	Binders	Ibico Hi-Tech Manual Binding System	100%
OFF-PA-10001883	Office Supplies	Fasteners	Alliance Super-Size Bands, Assorted Sizes	100%
OFF-PA-10000855	Office Supplies	Paper	Southworth 25% Cotton Granite Paper & Envelopes	100%
OFF-PA-10000205	Office Supplies	Paper	Xerox 225	100%
TEC-PH-10004039	Technology	Phones	Wireless Extenders Zboost Yx545 Soho Signal Booster	100%
TEC-PH-10004977	Technology	Phones	Ge 30524Ee4	100%
FUR-FU-10004664	Furniture	Furniture	Howard Miller 14-1/2" Diameter Chrome Round Wall Clock	100%
OFF-BI-10000108	Office Supplies	Binders	Acco Four Pocket Poly Ring Binder With Label Holder, Smoke, 1"	100%
OFF-AB-10004078	Office Supplies	Art	Newell 312	100%
TEC-AC-10001046	Technology	Accessories	Memorex Micro Travel Drive 8 Gb	100%
FUR-FU-10000965	Furniture	Furniture	Howard Miller 11-1/2" Diameter Ridgewood Wall Clock	100%
OFF-LA-10004172	Office Supplies	Labels	Avery 482	100%
FUR-FU-10000010	Furniture	Furniture	Dax Value U-Channel Document Frames, Easel Back	100%
FUR-BO-10000034	Furniture	Bookcases	O'Sullivan Elevations Bookcase, Cherry Finish	100%
FUR-CH-10000602	Furniture	Chairs	Global Highback Leather Tiller in Burgundy	100%
OFF-AB-10000514	Office Supplies	Art	4009 Highlighters By Sanford	100%
			Avery Metallic Poly Binders	100%

5. Checking Blank, Null and Error Values :

The dataset was checked for missing or invalid values to ensure that the data was suitable for further analysis.

Steps Followed :

1. Selected the required column.
2. Opened the column filter.
3. Examined the values displayed in the filter list.
4. Checked for **blank values**.
5. Checked for **empty values**.
6. Looked for **null values**.
7. Checked whether any **Error** values were present.
8. Reviewed the columns containing unwanted or missing values.
9. Corrected or handled the values wherever necessary.
10. Rechecked the columns after the cleaning process.

This step helped to improve the quality and reliability of the dataset.

Product Name	Sales
Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets...	100%
Gbc Standard Plastic Binding Systems Combs	100%
Avery 508	100%
Safco Boltless Steel Shelving	100%
Avery Hi-Liter Everbold Pen Style Fluorescent Highlighters, 4/Pack	100%
Global Deluxe High-Back Manager's Chair	100%
Rogers Handheld Barrel Pencil Sharpener	100%
Dixon Prang Watercolor Pencils, 10-Color Set With Brush	100%
Ibico Hi-Tech Manual Binding System	100%
Alliance Super-Size Bands, Assorted Sizes	100%
Southworth 25% Cotton Granite Paper & Envelopes	100%
Xerox 225	100%
Wireless Extenders Zboost Yx545 Soho Signal Booster	100%
Ge 30524Ee4	100%
Howard Miller 14-1/2" Diameter Chrome Round Wall Clock	100%
Acco Four Pocket Poly Ring Binder With Label Holder, Smoke, 1"	100%
Newell 312	100%
Memorex Micro Travel Drive 8 Gb	100%
Howard Miller 11-1/2" Diameter Ridgewood Wall Clock	100%
Avery 482	100%
Dax Value U-Channel Document Frames, Easel Back	100%
O'Sullivan Elevations Bookcase, Cherry Finish	100%
Global Highback Leather Tiller in Burgundy	100%
4009 Highlighters By Sanford	100%
Avery Metallic Poly Binders	100%

6. Performing Column Transformations

Column transformations were carried out to make the dataset easier to understand and use during analysis.

A. Renaming Columns :

Steps Followed :

1. Selected the required column.
2. Right-clicked on the column.
3. Selected **Rename**.
4. Entered the appropriate column name.
5. Pressed Enter.
6. Verified the updated column name.

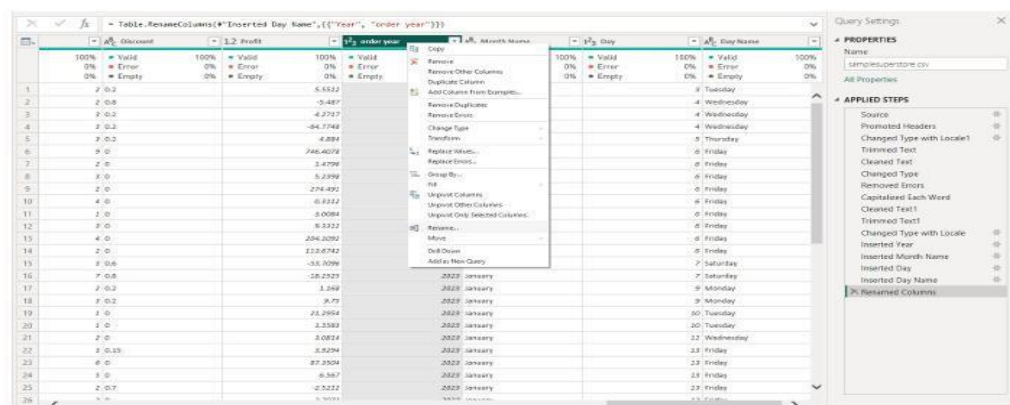
Renaming columns helped to make the dataset fields clear and meaningful.

B. Splitting Columns :

Where required, columns containing multiple pieces of information were divided into separate columns.

Steps Followed :

1. Selected the required column.
2. Opened the **Transform** tab.
3. Selected **Split Column**.
4. Selected the appropriate splitting option.
5. Defined the required delimiter or position.
6. Applied the transformation.
7. Checked the newly created columns.



7. Creating Date Attributes from Order Date :

The **Order Date** column was converted into the **Date** data type. From this column, additional date-related attributes were created to support time-based analysis.

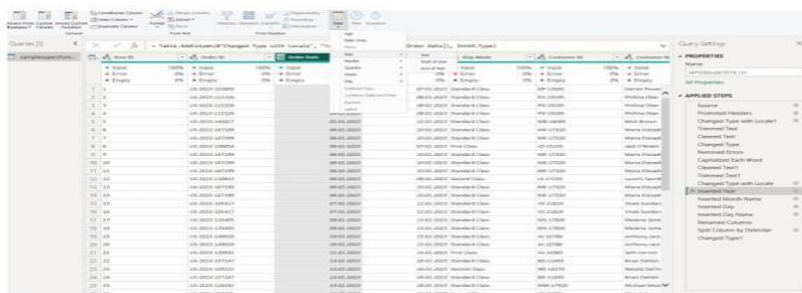
The following attributes were generated:

- Year
- Quarter
- Month
- Day
- Day of Week

A. Creating Year :

Steps Followed :

1. Selected the **Order Date** column.
2. Opened the **Add Column** tab.
3. Selected **Date**.
4. Selected **Year**.
5. Chose **Year** from the available option.
6. A new Year column was created.
7. Verified the generated values.

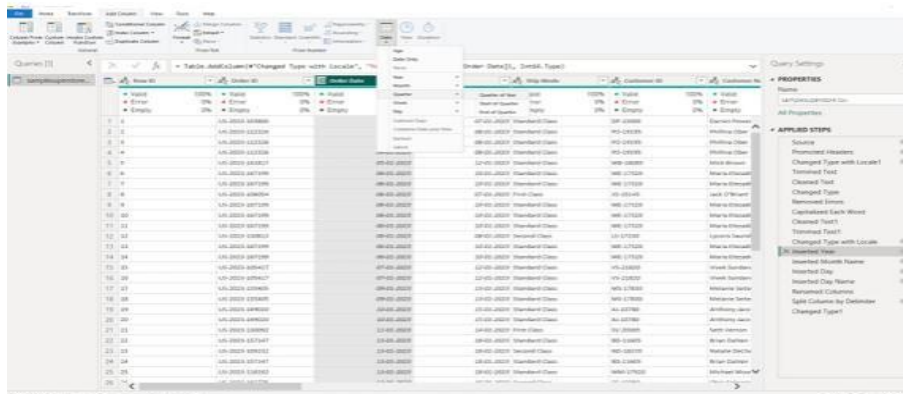


B. Creating Quarter :

Steps Followed :

1. Selected **Order Date**.
2. Clicked **Add Column**.
3. Selected **Date**.
4. Selected **Quarter**.

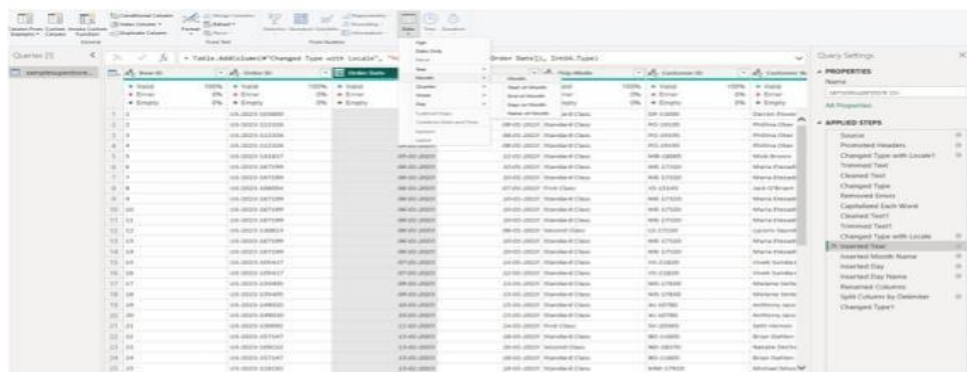
5. Chose **Quarter of Year**.
6. Checked the newly created Quarter column.



C. Creating Month :

Steps Followed :

1. Selected the **Order Date** field.
2. Went to **Add Column**.
3. Selected **Date**.
4. Selected **Month**.
5. Selected the required month option.
6. Verified the generated Month column.



D. Creating Day :

Steps Followed :

1. Selected the **Order Date** column.
2. Opened **Add Column**.
3. Selected **Date**.
4. Selected **Day**.
5. Chose the **Day** option.
6. Checked the newly generated Day column.

E. Creating Day of Week :

Steps Followed :

1. Selected the **Order Date** column.
 2. Opened the **Add Column** tab.
 3. Selected **Date**.
 4. Selected **Day**.
 5. Chose **Name of Day**.
 6. Verified the Day of Week values.
-

8. Final Data Validation :

After completing all the required cleaning and transformation operations, I performed a final review of the dataset to make sure that the data was accurate and properly prepared for analysis.

Final Checks :

- Verified all important data types.
- Confirmed the **Order Date** column was in Date format.
- Verified the **Year, Quarter, Month, Day, and Day of Week** columns.
- Checked for blank and empty values.
- Verified the presence of null values.
- Checked the dataset for any error values.
- Reviewed the cleaned categorical text fields.
- Confirmed that the required columns were renamed correctly.
- Verified the transformed and newly created columns.
- Reviewed the complete dataset before applying the changes.

This final validation ensured that the dataset was clean, consistent, and ready to be loaded into Power BI.

Queries [1] Table.AddColumn('Changed Type with Locale', 'Year', each Date.Year([Order Date]), Int64.Type)

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	1	US-2023-101800	09-01-2023	07-01-2023	Standard Class	OP-13000	Barbara Power
2	2	US-2023-111316	04-01-2023	08-01-2023	Standard Class	PO-13135	Phillina Ober
3	3	US-2023-111316	04-01-2023	08-01-2023	Standard Class	PO-13135	Phillina Ober
4	4	US-2023-111316	04-01-2023	08-01-2023	Standard Class	PO-13135	Phillina Ober
5	5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MR-18085	Mick Brown
6	6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
7	7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
8	8	US-2023-106054	08-01-2023	07-01-2023	First Class	JO-15145	Jack O'Brian
9	9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
10	10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
11	11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
12	12	US-2023-150813	06-01-2023	08-01-2023	Standard Class	LS-17330	Lynette Swand
13	13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
14	14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Escada
15	15	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Viola Sundara
16	16	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Viola Sundara
17	17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
18	18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
19	19	US-2023-140000	10-01-2023	15-01-2023	Standard Class	AI-10780	Anthony Jack
20	20	US-2023-140000	10-01-2023	15-01-2023	Standard Class	AI-10780	Anthony Jack
21	21	US-2023-130062	11-01-2023	14-01-2023	First Class	SV-20365	Seth Verman
22	22	US-2023-157147	12-01-2023	18-01-2023	Standard Class	BD-11805	Brian Dehen
23	23	US-2023-180312	11-01-2023	14-01-2023	Standard Class	MS-18170	Maria Dehen
24	24	US-2023-157147	12-01-2023	18-01-2023	Standard Class	BD-11805	Brian Dehen
25	25	US-2023-118192	12-01-2023	18-01-2023	Standard Class	MM-17120	Michael Moe
26	26	US-2023-118192	12-01-2023	18-01-2023	Standard Class	MM-17120	Michael Moe

Query Settings

PROPERTIES

Name: samplestore.cdv

APPLIED STEPS

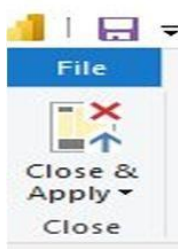
- Source
- Promoted Headers
- Changed Type with Locale1
- Trimmed Text
- Cleaned Text
- Changed Type
- Removed Errors
- Capitalized Each Word
- Cleaned Text1
- Trimmed Text1
- Changed Type with Locale
- Inserted Year**
- Inserted Month Name
- Inserted Day
- Inserted Day Name
- Renamed Columns
- Split Column by Delimiter
- Changed Type1

9. Close and Apply :

Once all the data cleaning and transformation activities were completed, I applied the changes made in Power Query Editor to the Power BI data model.

Steps Followed :

1. Reviewed the complete transformed dataset in **Power Query Editor**.
2. Checked the final changes and transformations.
3. Opened the **Home** tab.
4. Selected **Close & Apply**.
5. Power BI processed and applied all the transformations.
6. The cleaned dataset was successfully loaded into the **Power BI data model**.



Final Result:

The **Superstore Sales Dataset** was successfully imported and prepared using **Microsoft Power BI and Power Query Editor**. The required data types were verified and corrected, and text fields were standardized using **Trim** and **Clean** operations.

The dataset was also checked for **blank, empty, null, and error values**. Required column transformations, including **Rename** and **Split**, were performed. The **Order Date** field was

used to generate useful time-based attributes such as **Year, Quarter, Month, Day, and Day of Week**.

After completing all the transformations, the final dataset was validated and loaded successfully into the Power BI data model using **Close & Apply**.

Conclusion :

In this task, I successfully performed the basic data preparation process using **Microsoft Power BI and Power Query Editor**. I learned how to import a CSV dataset, inspect its structure, verify data types, and identify missing or incorrect values.

I also learned how to clean text data using **Trim and Clean**, perform column transformations, and create useful date attributes from the **Order Date** column. Finally, I validated the transformed dataset and used **Close & Apply** to load the prepared data into Power BI.

This activity helped me gain practical knowledge of **data cleaning, transformation, and preparation**, which are important steps before creating meaningful reports and dashboards in Power BI.